

Task 1:- Scenario Analysis

1. To maintain neutrality, we have long option positions and short shares of the underlying stock. If the delta (Δ) of the long options is x and the number of shares represented is y , then the total positive delta of the options is xy . This is balanced by the short shares which have a delta of $-xy$, making the net portfolio delta equal to 0. On Wednesday, the company announced a major restructuring plan and the stock price increased by 18% overnight, i.e. from \$150 to \$177. The long options became much more valuable and their delta increased from x to $(x + z)$. However, our short stock position remained the same overnight because the market was closed and we could not rebalance the hedge. Therefore, the positive delta from the options became $(x + z)y$ while the short stock delta remained $-xy$. The net portfolio delta became $(x + z)y - xy = zy$, which is greater than zero. Hence, the portfolio is no longer delta-neutral at the Thursday morning open. The Greek responsible for this change is Gamma (Γ). Gamma measures how much the delta of an option changes when the price of the underlying stock changes. Since the stock price increased significantly overnight, the delta of the long options also increased, causing the portfolio to lose its delta-neutral position.
2. We were holding a Long Gamma position because we were holding long options. Since long options have positive gamma, the delta of the position changed significantly as the underlying price moved. As the stock increased sharply, the delta of the long options increased rapidly. Because we were not able to rebalance the hedge overnight, this change in delta resulted in a larger positive exposure, and the option position benefited from the upward move in the underlying. Therefore, the overnight price gap worked in our favour from a gamma perspective, as long gamma positions benefit from large movements in the underlying asset.
3. With high IV, premiums are high, so as the IV crash happened due to restructuring, the premiums became low. The greek is Vega. It measures how much an option's premium changes for every 1% shift in the underlying asset's Implied Volatility (IV).